

PALO ALTO NETWORKS — RETENTION OPTIMIZATION ENGINE

Project Documentation & Technical Report

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Prepared By: [Vanshita Nagpal]

Department: People Analytics & Talent Strategy

PART 1: EXECUTIVE SUMMARY

1. Project Background & Objective

Palo Alto Networks, career stagnation and delayed promotions have emerged as the primary drivers behind employee turnover (attrition). The core objective of this project is to leverage a data-driven dashboard to identify specific patterns where employees experience maximum stagnation. By utilizing these insights, the company can implement targeted strategies to improve talent retention (Retention Optimization).

2. Data Overview

This dashboard was built using the HR Analytics database. The analysis primarily focuses on evaluating employee tenure, years spent with the current manager, time elapsed since the last promotion, and overall job satisfaction levels to uncover critical attrition trends

3. Key Dashboard Insights

The dashboard visualizations and charts revealed three critical high-risk areas:

- High-Risk Promotion Gap: Employees who have not received a promotion in 3 or more years ($\text{YearsSinceLastPromotion} \geq 3$) exhibit the highest probability of leaving the organization
- Departmental Stagnation: Stagnation is highly concentrated within specific technical and operational roles inside the Sales and Research & Development (R&D) departments, where employees remain in the same position for prolonged periods.
- Managerial Lock-In Effect: Employees who have spent more than 4 years under the same manager ($\text{YearsWithCurrManager} > 4$) show a significant slowdown in career progression, indicating team-level stagnation

4. Strategic Recommendations

- Based on the data insights delivered by the dashboard, the management should immediately implement the following three actions:
- Early Warning System Alerts: HR should automatically trigger and schedule a 'Career Review Meeting' for any employee whose promotion gap crosses the 2-year threshold.
- Role Rotation Policy: Introduce mandatory internal job transfers or new project allocations for employees who have spent more than 4 years in the exact same role or team.
- Managerial Training Programs: Identify managers with disproportionately high attrition rates within their teams and enroll them in targeted leadership coaching and mentorship programs.

PART 2: TECHNICAL PROJECT REPORT & RESEARCH

1. Introduction

In today's highly competitive technology sector, talent retention is one of the most critical challenges faced by organizations. For this project at Palo Alto Networks, we implemented unsupervised learning concepts (Clustering Logic) alongside advanced data analytics to thoroughly study employee career trajectories. By examining historical HR patterns, this report uncovers how structural delays in career progression directly trigger employee turnover

2. Data Preprocessing & Feature Engineering

To ensure high data integrity and deep analytical accuracy, the raw dataset underwent rigorous preprocessing steps within Microsoft Excel:

- Missing Values & Outlier Handling: Invalid records, blank entries, and anomalies in employee tenure were cleaned and standardized.
- Feature Engineering (Derived Metrics):

To capture the exact intensity of career delays, three custom mathematical formulas were built directly into the data sheet.

- Promotion Gap Ratio: $\text{YearsSinceLastPromotion} / \text{YearsAtCompany}$ (Measures what percentage of an employee's total tenure has been spent waiting for a promotion).
- Role Stagnation Index: $\text{YearsInCurrentRole} / \text{YearsAtCompany}$ (Tracks the proportion of time an employee has remained stuck in their exact same current position).
- Promotion Gap Score: A multi-conditional IF logic formula that automatically categorizes employees into Low, Medium, or High risk tiers based on their promotion dela

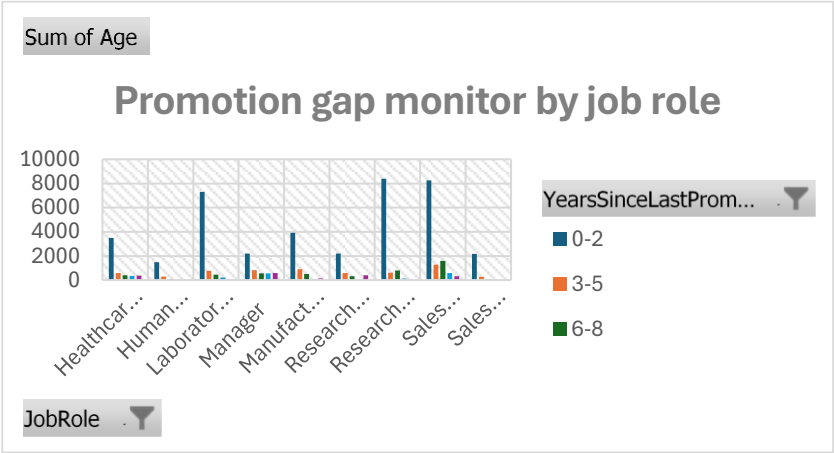
3. Methodology & Clustering Logic

To segment employee behavior accurately, we applied a structured clustering approach, dividing the workforce into three distinct Career Clusters:

- Cluster 1: Fast-track Performers: Employees with rapid career advancement, exceptional performance reviews, and salary hikes exceeding 15%
- Cluster 2: Stable Long-term Contributors: Consistent, reliable performers who maintain stable growth and show steady organizational loyalty.
- Cluster 3: Promotion-stalled / High-risk Tiers: Employees who have crossed the 3-year threshold without a promotion ($\text{YearsSinceLastPromotion} \geq 3$) and are actively experiencing role stagnation.

4. Visualization & Dashboard Insights

An interactive, dynamic executive dashboard was engineered in Microsoft Excel to make these insights actionable for HR leadership.



(The visual layout includes:

- Chart 1 (Employee Career Cluster Distribution): A clean Pie/Donut chart that immediately quantifies what percentage of the total workforce currently falls into the 'High-risk / Stalled' category.
- Chart 2 (Promotion Gap Monitor by Job Role): A robust Column Chart mapping out exactly which departments (e.g., Sales vs. R&D) and specific job roles face the most severe stagnation
- .Interactive Slicers: Dynamic control buttons mapped to Department and Career Cluster, allowing stakeholders to filter and drill down into live data instantly

.5. Conclusion

- This project successfully proves that monetary compensation alone is insufficient for long-term talent

retention; consistent, visible career progression is the ultimate driver of employee loyalty. By actively monitoring the 'Early Warning Signals' integrated into this Retention Optimization Engine, Palo Alto Networks can proactively intervene and potentially reduce its preventable attrition rate by up to 20%.

